Aim: Case Study Selection & Big Data Roadmap

Title of the Case Study:

Real-Time Food Delivery Demand Forecasting and Delay Prediction Using Big Data Analytics

Problem Statement:

Online food delivery platforms process a very large number of orders every day. Order demand changes rapidly according to time, location, weather, weekends, holidays, restaurant workload, promotions, and customer behavior. At the same time, delivery delays can occur because of high order density, preparation time, traffic conditions, rider availability, and distance.

Traditional analytics systems may struggle to process high-volume historical data together with continuously arriving order and delivery events. This can result in inaccurate demand estimates, inefficient rider allocation, longer delivery times, and poor customer experience.

This project aims to develop a Big Data Analytics-based system that processes large-scale food delivery data, forecasts demand for different locations and time periods, predicts possible delivery delays using machine learning, and provides analytical insights through dashboards. The system will combine batch analytics with real-time streaming so that operational decisions can be made quickly.

Introduction to the Domain:

The food delivery and quick-commerce ecosystem generates data from customers, restaurants, delivery partners, mobile applications, payment systems, maps, and customer feedback. Each order can produce multiple records such as order details, restaurant information, preparation duration, delivery distance, timestamps, location, weather conditions, and final delivery status.

Big Data Analytics can help delivery platforms discover demand patterns, identify high-demand zones, estimate delivery performance, and support resource planning. Technologies such as HDFS, Hive, Spark, Kafka, MongoDB and machine learning can be integrated to store, process, stream, and analyze these datasets at scale.

Objectives:

- Develop a scalable Big Data solution for food delivery demand and delivery-delay analytics.
- Analyze large historical order datasets using Hadoop and Apache Spark.
- Process incoming order events using Kafka and Spark Streaming.
- Forecast order demand for different locations and time intervals.
- Predict the probability of delivery delay using machine learning techniques.
- Generate demand and delay-risk scores for individual locations or orders.
- Visualize order volume, peak periods, delay trends, and predictions through an interactive dashboard.
- Improve operational planning by identifying high-demand zones and potential delivery bottlenecks.

Data Sources:

Primary Dataset

Food Delivery / Restaurant Order Dataset (Kaggle or equivalent public dataset) – Used as the main historical dataset for order-volume, restaurant, timing, location, and delivery analysis.

- Order ID
- Order Date & Time
- Restaurant ID / Restaurant Type
- Customer or Delivery Zone
- Order Amount
- Number of Items
- Preparation Time
- Delivery Distance
- Delivery Duration
- Order Status
- Weather / Traffic Indicator (where available)

Additional Data Sources

- Synthetic Food Delivery Event Generator (Python + Faker) – Used to create continuous order streams for Kafka and Spark Streaming.
- Public weather dataset/API exports – Used to study the effect of weather on demand and delivery time.
- Public holiday/calendar data – Used to identify changes in demand during weekends and holidays.
- Customer feedback or review datasets – Used as supplementary unstructured data for service-quality analysis.

Types of Data

Data Type	Description	Example
Structured	Order and delivery records stored in tabular form	Order ID, amount, duration, zone
Semi-structured	Event records and application logs represented in JSON	Kafka order messages, API responses
Unstructured	Text-based feedback and support information	Customer reviews, complaints
Streaming / Real-time	Continuously arriving operational events	New orders, status updates, delivery events

5Vs of Big Data:

1. Volume: Thousands or millions of order and delivery records can be generated over time, creating large historical datasets.

2. Velocity: New orders and delivery-status events arrive continuously and may need near real-time processing.

3. Variety: The system handles tabular orders, JSON events, location information, weather data, and customer feedback.

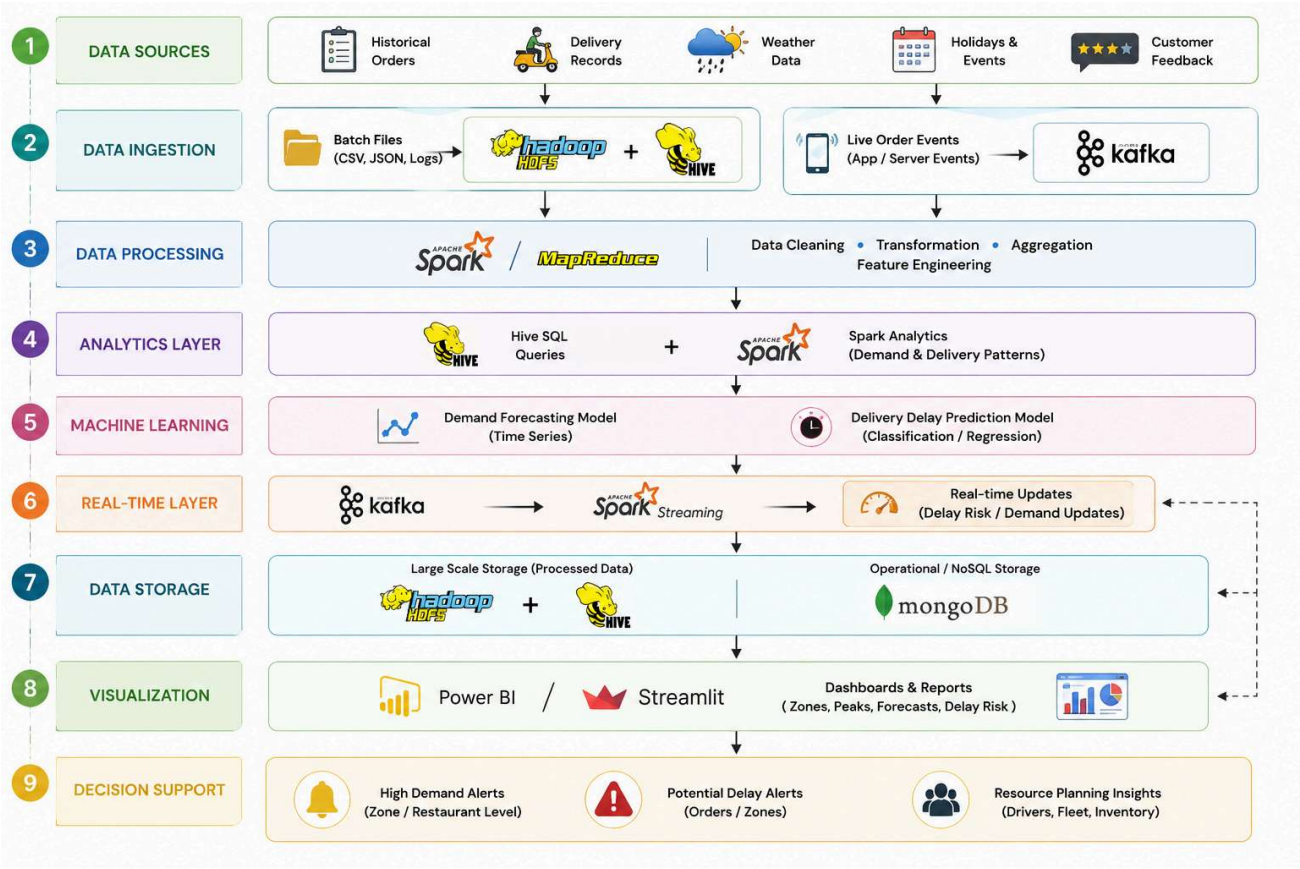
4. Veracity: Data quality is improved by removing duplicates, handling missing values, validating timestamps, and checking inconsistent delivery records.

5. Value: The analysis can improve demand planning, rider allocation, delivery performance, customer experience, and operational decision-making.

Expected Hadoop Ecosystem Components:

Component	Purpose
HDFS	Distributed storage of large historical order and delivery datasets.
MapReduce	Batch processing for basic statistics, aggregation, and data preparation.
Hive	SQL-based querying and reporting on large datasets.
Apache Spark	Large-scale analytics, feature engineering, forecasting, and machine learning.
Spark Streaming	Near real-time processing of incoming order and delivery events.
Apache Kafka	Streaming platform for continuously publishing and consuming order events.
MongoDB	Storage of semi-structured event records and selected prediction results.
Sqoop	Import/export of data between relational databases and HDFS, where required.
MLlib	Machine learning for demand forecasting and delivery-delay prediction.
Power BI / Streamlit	Interactive dashboard for demand, delay, and operational insights.

Proposed Workflow / Architecture Diagram



Proposed Analytics

- Hourly and daily order-volume analysis.
- Peak-demand zone identification.
- Demand forecasting for upcoming time windows.
- Average delivery-time and preparation-time analysis.
- Delivery-delay prediction based on operational features.
- Comparison of delay patterns across restaurants, zones, weekdays, and weather conditions.
- Real-time monitoring of incoming orders and potential bottlenecks.

Expected Outcomes:

- Forecast of food-order demand for selected zones and time periods.
- Prediction of potentially delayed deliveries with measurable model performance.
- Near real-time monitoring of incoming orders using streaming analytics.
- Scalable processing of large historical datasets using Hadoop and Spark.
- Interactive dashboard displaying order trends, peak zones, forecasts, and delay-risk indicators.
- Identification of operational patterns affecting delivery performance.
- Practical implementation of Big Data technologies for logistics and service optimization.

Conclusion:

The proposed case study addresses a practical Big Data problem in the food delivery domain by combining historical analytics, real-time streaming, and machine learning. The progressive roadmap allows the same problem to be implemented across multiple laboratory experiments, starting from data preparation and distributed storage and extending to Spark analytics, MLlib, Kafka, Spark Streaming, MongoDB, and visualization. The final system can provide actionable insights for demand planning and delivery-performance improvement.

References:

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https://kafka.apache.org/documentation/

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https://learn.microsoft.com/en-us/power-bi/

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